

CG.3.403 Behavioral Research: Statistical Methods

Course Information

Instructor Information

Instructors: Dr. Vishnu Sreekumar and Dr. Bapi Raju
with guest lectures by Dr. Bhaktee Dongaonkar

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Day/Time: Mondays and Thursdays: 11:40 am – 1:05 pm

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Course Information

Course Description: Students will be introduced to the different statistical methods employed in the analysis of behavioral and other data. The material will be delivered as a combination of lectures and practical sessions. In the practical sessions, students will be provided with data and code snippets to help them practice the concepts taught in the lectures. They will also receive regular problem sets/assignments/projects. We will primarily rely on R for statistical analysis but may also use other tools as deemed appropriate for the material being covered.

Credits: 4

L-T-P: 3-0-1 (L = lecture hours, T = tutorial hours, P = practical hours)

Prerequisite: None

Textbook & Course Materials

Recommended Texts & Other Readings: Learning Statistics with R by Danielle Navarro: <https://learningstatisticswithr.com/lr-0.6.pdf>

Lecture slides and supplementary reading materials (journal articles, books/book chapters, online resources) will be uploaded on the course page on Moodle.

Course Technology Requirements

Laptop to be brought to class for practical sessions. Please ensure enough battery charge and internet access to submit your work on Moodle by the end of class time for the practical sessions.

Course Structure

This course will be delivered as in-person lectures and practical sessions.

Student Expectations

In this course you will be expected to complete the following types of tasks.

- **Read the syllabus!**
- Meet deadlines
- Attend in-person classes
- Participate in practical sessions every week
- Submit your practical class-work every week
- Communicate via email
- Complete basic internet searches
- Download and upload documents to the course site on Moodle
- read documents online
- view online videos
- participate in class discussions
- complete quizzes/tests
- upload documents to a Dropbox/Moodle

Expected Instructor/TA Response Times

- We will attempt to respond to student emails within 24 hours. If you have not received a reply from us within 24 hours, please resend your email. Please email both of us to maximize the probability of a quick response.
 - ***If you have a general course question (not confidential or personal in nature), please post it to the Course Q&A Discussion Forum found on the course homepage on Moodle. We will post answers to all general questions there so that all students can view them. Students are encouraged to answer each other's questions too.
- We will attempt to reply to and assess student discussion posts within 48 hours.

Course Outcomes (COs)

After successful completion of this course, students will be able to:

- CO-1: develop an understanding of various experimental designs
- CO-2: recognize and employ appropriate statistical packages to analyze data
- CO-3: apply appropriate parametric and non-parametric analyses techniques
- CO-4: perform exploratory data analysis and examine intrinsic relationships between variables
- CO-5: reflect and draw appropriate inferences post analyses
- CO-6: create custom code by adapting exploratory and confirmatory analyses techniques

You will meet the outcomes listed above through a combination of the following activities in this course:

- Attend lectures and participate in class discussions (CO-1, CO-2, CO-3, CO-4, CO-5)
- Quiz 1, Quiz 2, and Endsem (CO-1, CO-2, CO-5)
- Complete problem sets, assignments, and mini-project(s) (CO-2, CO-3, CO-4, CO-5, CO-6)

Mapping of Course Outcomes to Program Objectives: (1 – Lowest, 2 – Medium, 3 – Highest, or a '-' dash mark if not at all relevant). Program outcomes are posted at https://iiitaphyd-my.sharepoint.com/:w:/r/personal/dyacad_iiit_ac_in/Documents/NBA-2020-21/Course%20Content/IIIT-CSE-ECE.docx?d=w111f0effcaea41b3a4d1e8a3fbc6332d&csf=1&web=1&e=z1Khby

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	3	1	1	1	2	1	1	1	1	-	-	-	-	-	-
CO2	2	2	2	3	3	2	1	1	3	1	-	-	-	-	-	-
CO3	2	3	2	3	3	3	1	1	3	1	-	-	-	-	-	-
CO4	2	3	2	3	3	3	1	1	3	1	-	-	-	-	-	-
CO5	1	1	1	1	1	3	1	1	2	1	-	-	-	-	-	-
CO6	2	2	3	3	3	1	1	1	2	2	-	2	1	1	2	3

Tentative Schedule

You can find a tentative schedule [here](#) but it is subject to change as we proceed through the semester.

- Week 1: Introduction: why do statistics, and basics of research design
- Week 2: Installing R and basics of R
- Week 3: Descriptive statistics and visualizing data
- Week 4: Organizing, sorting, merging, and reading data in R
- Week 5: Probability and sampling distributions
- Week 6: Null hypothesis testing
- Week 7: Categorical data analysis, comparing two means, data reduction
- Week 8: Linear regression, regression diagnostics, and related hypothesis tests
- Week 9: One-way ANOVA
- Week 10: Factorial ANOVA
- Week 11: Generalized linear models (GLMs)
- Week 12: Bayesian statistics and inference

Grading Policies

Graded Course Activities

Description	Percentage
Assignments	20
Project	30
2 Quizzes (10% each)	20
End sem exam	30
Total	100
Bonus	3

No midsem (project progress, report 1 = midsem).

Quizzes and end-sem exams

Quizzes will mostly have multiple-choice questions, and the end-sem exam will be a mix of multiple-choice and descriptive questions.

In-class problem sets

We will provide sample datasets and problems to work on in class, and code snippets to facilitate them. You are highly encouraged to use Google Search during the practical sessions when analyzing data using R. Students are expected to attend all the practical sessions and submit their answers at the end of the class. This is not a graded component. **However, those of you who submit all in-class assignments within the stipulated time (usually by the end of class) will receive upto a 3% bump in your final percentage which may help you cross a grade cutoff if you happen to be close to a cutoff.** If you submit all in-class problem sets **having attempted the questions properly**, you will receive the full 3% bonus. If you miss even one, you will not get this bonus. An additional component of this 3% bonus is the first project presentation. We will randomly call on teams to present. If you are absent or unprepared to present when called, you will not be eligible for the bonus.

Because of this provision, we will not entertain any requests to bump up percentages at the end of the course if you are close to a cutoff. Those who attend classes will benefit from this bonus.

Assignments (20% weightage)

Students will have to submit code and solutions to some take-home assignments. Plagiarism will result in 0 marks and the students will also be reported to the administration. If you have queries about what constitutes plagiarism, please read: <https://mrwachs.wordpress.com/2019/11/25/what-is-plagiarism-in-computer-science/>, <https://clas.iusb.edu/computer-science-informatics/prospective-students/plagiarism.html>, and also contact the instructors by email with your questions.

If you are stuck on a problem, brain-storming with a friend or group of friends is encouraged, but not looking at their code and copying their code and logic. The latter will result in 0 marks awarded for the assignment, and a letter grade penalty for each further infraction.

Quizzes and the endsem may have questions from the in-class and take-home assignments. These questions will test the logic of the code you wrote for the assignments. People who use AI for their assignments without applying their own minds will not be able to perform well on the exams. **If we suspect blind AI use, we reserve the right to give a random viva to assess your understanding of your own submission.**

Mini Project (60 marks, 30% weightage)

You will first participate in human behavioral experiments (at least 1.5-2 hrs worth of experiments). Then, **by Feb 10**, you will be provided with anonymized data from one of the experiments you took part in along with some sample questions (non-exhaustive, you are supposed to come up with more questions on your own) you can answer **by applying statistical methods** on those datasets. **Note that machine learning methods will not be accepted.** You have to primarily apply the methods taught in class with a few additional methods if needed. However, **using chatGPT to come up with sophisticated analyses that no one understands or if you are unable to properly interpret the results of your analyses when asked (again indicating that you copy-pasted the output of AI without making an effort to really understand it) will attract a big 50% penalty.** Therefore, we reserve the right to give a viva to any person/team that engages in such extreme genAI assistance with very little understanding of the statistical methods, their assumptions, their applicability to your data, and the appropriate interpretation of the results.

The graded components are:

1. Experiment participation – 10 marks (experiments will have vigilance checks, so you will lose 10 marks if you do not pay attention and do the task to the best of your ability).
2. Report 1 – presenting visualization of variables in the data, descriptive statistics, and any other preliminary analysis results (both descriptive and inferential – with plots and statistics) - 10 marks.
3. Presentation of report 1 – randomly selected teams (1 per experiment/dataset) who will receive a 1% bonus.
4. Report 2 – update of the initial report with additional inferential statistical analyses using advanced methods such as ANOVA, regression, and GLM – 20 marks.
5. Final poster presentation (instead of class presentations); all teams - 10 marks.

Report 1 (~5-6 pages, single-spaced, font size 12) to be submitted by **10 pm Feb 21, 2025. You will forfeit the 10 marks associated with report 1 if you miss this deadline.**

Report 2 (~10-12 pages, single-spaced, font size 12) **to be submitted by 10 pm April 20, 2025. You will forfeit the 20 marks associated with report 2 if you miss this deadline.**

Both reports will be graded according to the same scheme (but half the marks for report 1):

1. Introduction and explanation of the dataset, some background about the domain (you should do some research to understand the domain) – 2 marks
2. Methods (the analysis you did must be clear from the methods section – this is where you describe HOW you analyzed the data) – 6 marks

3. Results (make good figures, use statistics to make your points about what you present in the figures – this is where you describe WHAT you found by doing the analysis) – 8 marks
4. Conclusion from the analyses above and plans for the next steps/future directions – 2 marks
5. Formatting, citations, references – 2 marks

Total 20 marks (10 for report 1)

Final Poster Presentation – April 23rd, 2026

Due to the big class size, we will not be able to have project presentations in class. Instead, we will do a poster session with all teams presenting simultaneously. We will invite faculty to attend your presentations. You will also submit a PDF of your poster by 10 pm on April 22, 2026.

Participation

Students are expected to participate in all class activities as listed on the course calendar. *Failure to participate will result in students being unable to complete the in-class problem sets, assignments, and project satisfactorily and any such resulting effect on the final grade is entirely the student's responsibility.*

Complete Assignments

All assignments for this course will be submitted electronically through the course page on Moodle unless otherwise instructed.

Assignments must be submitted by the given deadline or special permission must be requested from instructor *before the due date*. Extensions will not be given beyond the next assignment except under extreme circumstances.

Late or missing assignments will affect the student's grade.

Late Work Policy

Be sure to pay close attention to deadlines—there will be no make-up assignments or quizzes, or late work accepted without a serious and compelling reason and instructor approval.

Viewing Grades on Moodle

Points you receive for graded activities will be posted to the course page on Moodle. Click on the Grades link to view your points.

Grading

Relative grading will be followed in this course. Final scores will be rounded up and graded accordingly. The bonus will be applied AFTER grade cutoffs are decided, so that you can take full advantage of the bonus.

Grading – BONUS policy

The bonus will be applied to your final percentage before determining your grade. However, the cutoffs will be decided prior to applying the bonus so that you can potentially benefit from the bonus.

Course Policies

Build Rapport

If you find that you have any trouble keeping up with assignments or other aspects of the course, make sure you let your instructor know as early as possible. As you will find, building rapport and effective relationships are key to becoming an effective professional. Make sure that you are proactive in informing your instructor when difficulties arise during the semester so that we can help you find a solution.

Inform Your Instructor of Any Accommodations Needed

If you have a documented disability and wish to discuss academic accommodations, please contact your instructors as soon as possible.

Statement of Policy

The instructors of this course will modify requirements as necessary to ensure that they do not discriminate against qualified students with disabilities. The modifications should not affect the substance of educational programs or compromise academic standards; nor should they intrude upon academic freedom. Examinations or other procedures used for evaluating students' academic achievements may be adapted. The results of such an evaluation must demonstrate the student's achievement in the academic activity, rather than describe his/her disability.

If modifications are required due to a disability, please inform the instructor.

Commit to Integrity

As a student in this course (and at IIIT Hyderabad) you are expected to maintain high degrees of professionalism, commitment to active

learning and participation in this class and also integrity in your behavior in and out of the classroom.

IIIT Hyderabad Academic Honesty Policy & Procedures

Student Academic Disciplinary Procedures

- (1) Academic misconduct is an act in which a student:
 - (a) Seeks to claim credit for the work or efforts of another without authorization or citation;
 - (b) Uses unauthorized materials or fabricated data in any academic exercise;
 - (c) Forges or falsifies academic documents or records;
 - (d) Intentionally impedes or damages the academic work of others;
 - (e) Engages in conduct aimed at making false representation of a student's academic performance; or
 - (f) Assists other students in any of these acts.

- (2) Examples of academic misconduct include, but are not limited to: cheating on an examination; collaborating with others in work to be presented, contrary to the stated rules of the course; submitting a paper or assignment as one's own work when a part or all of the paper or assignment is the work of another; submitting a paper or assignment that contains ideas or research of others without appropriately identifying the sources of those ideas; stealing examinations or course materials; submitting, if contrary to the rules of a course, work previously presented in another course; tampering with the laboratory experiment or computer program of another student; knowingly and intentionally assisting another student in any of the above, including assistance in an arrangement whereby any work, classroom performance, examination or other activity is submitted or performed by a person other than the student under whose name the work is submitted or performed.

We will be using plagiarism detection software. Please do not copy-paste from other papers. If you use direct quotes, you have to use the quotation marks "xyz" and cite your source: e.g. (Johnson & Johnson, 1988, p. 5). Please use APA format.

If plagiarism or academic misconduct of any kind is detected, we will award 0 marks for the assignment/exam/project as applicable. If plagiarism is detected a second time in another assignment/project

write-up, then one letter grade will be deducted from the final grade (e.g.¹ if you get a B, that will be changed to C) and you will be reported to the appropriate authorities for further disciplinary action.

Note: This syllabus was adapted from a template provided at www.uwsp.edu